Generative AI & LLMs

Lecture: Prompt Engineering & LLM inference parameters, Huggingface Transformers Library

Dr. Soharab Hossain Shaikh
Professor, CSE, SoET
BML Munjal University

1

Agenda

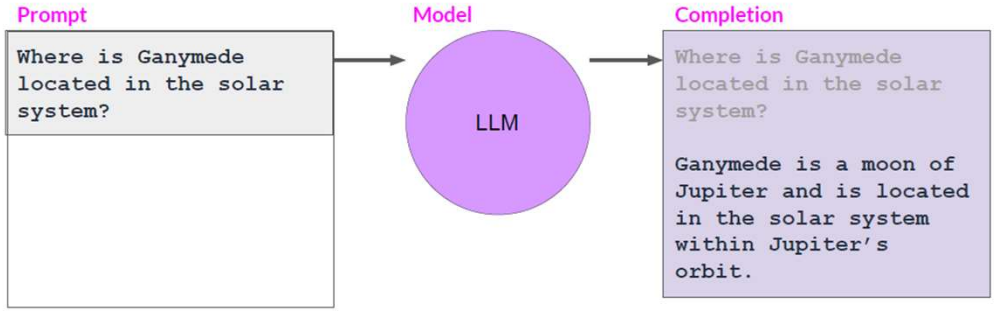
- Prompt & Completion
- In-context Learning (ICL) – zero-shot, one-shot- few-shot
- LLM Parameters – temperature, top_k, top_p
- Huggingface Transformers Library

2

Prompting & Prompt Engineering

3

Prompting and prompt engineering



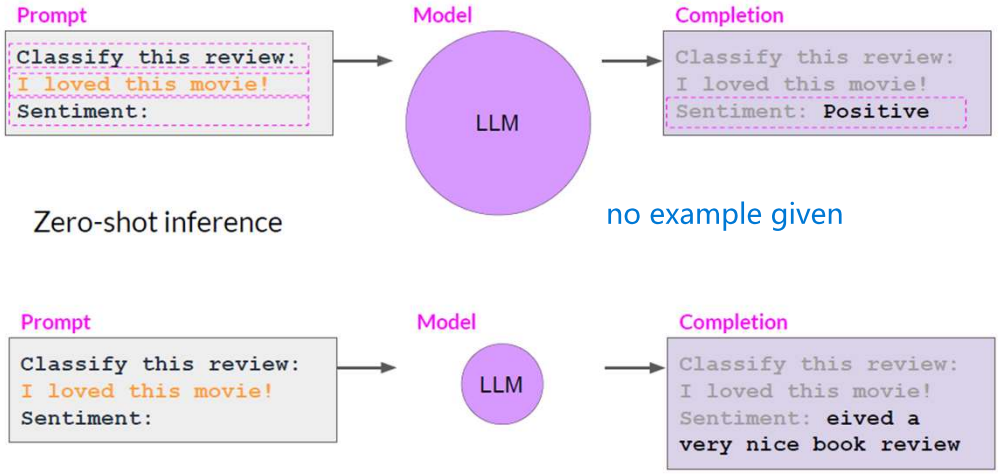
Context window: typically a few thousand words

LLM o/p- completion

Slide Credits: DeepLearning.ai course: Generative AI and LLMs with AWS
https://www.deeplearning.ai/courses/generative-ai-with-llms/

4

In-context learning (ICL) - zero shot inference

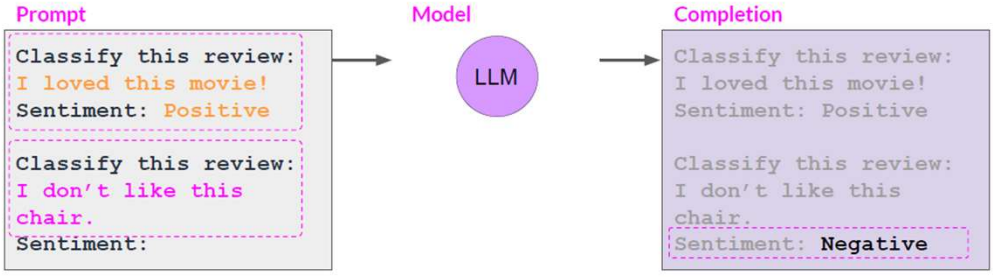


Zero-shot inference

no example given

5

In-context learning (ICL) - one shot inference



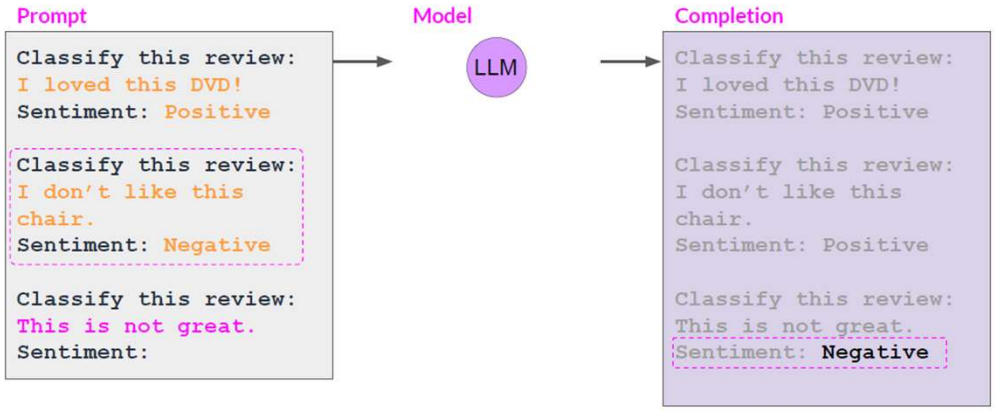
One-shot inference

1 example given

6

many examples

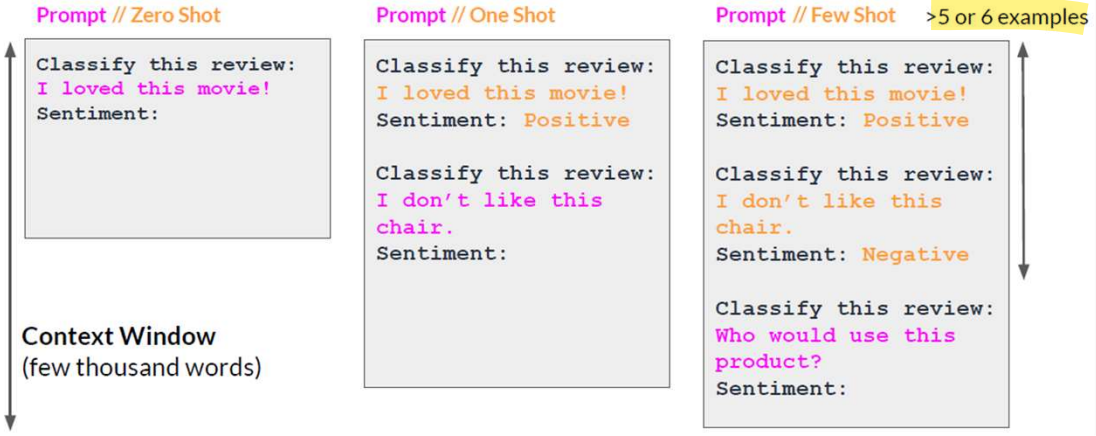
In-context learning (ICL) - few shot inference



improves performance

7

Summary of in-context learning (ICL)

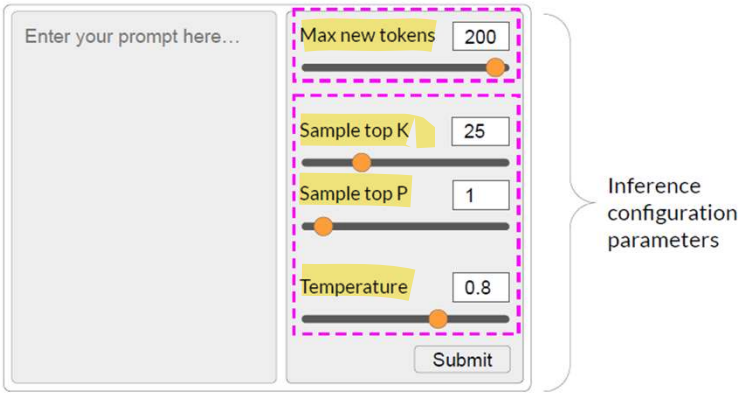


8

Generative Configurations Parameters for Inference

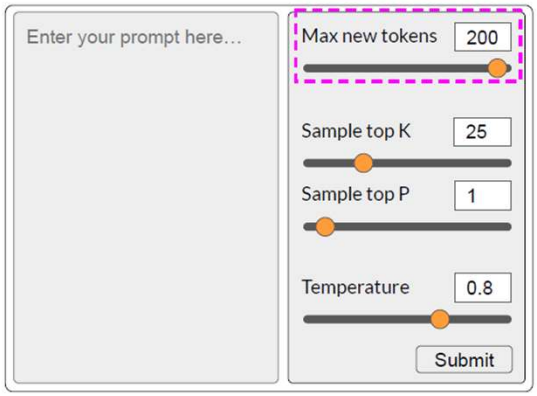
9

Generative configuration - inference parameters



10

Generative configuration - max new tokens

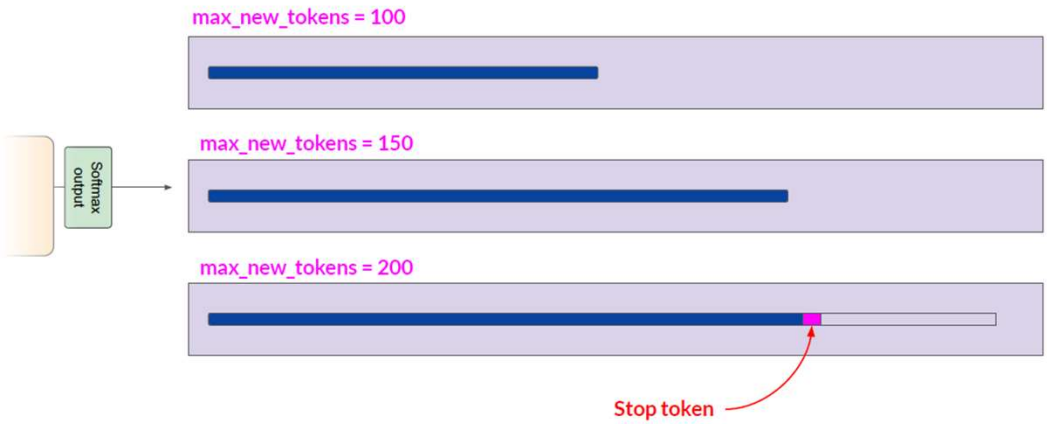


Max new tokens

11

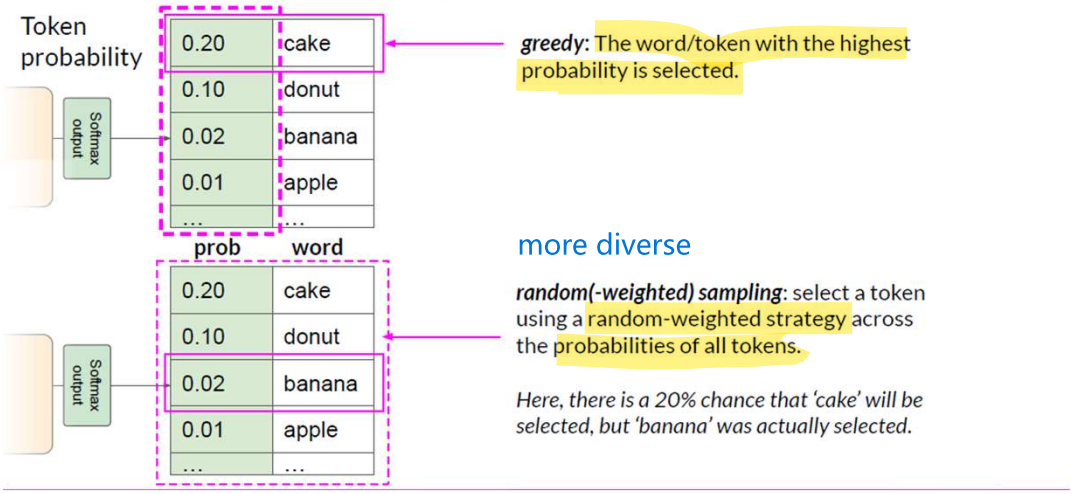
max length of o/p

Generative config - max new tokens



12

Generative config - greedy vs. random sampling



greedy: The word/token with the highest probability is selected.

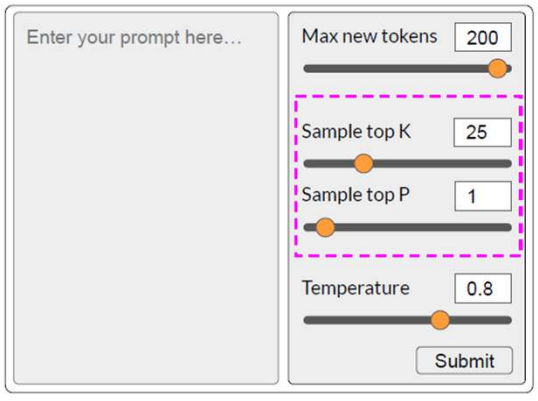
more diverse

random(-weighted) sampling: select a token using a random-weighted strategy across the probabilities of all tokens.

Here, there is a 20% chance that 'cake' will be selected, but 'banana' was actually selected.

13

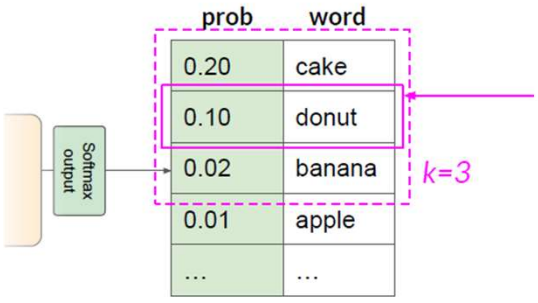
Generative configuration - top-k and top-p



Top-k and top-p sampling

14

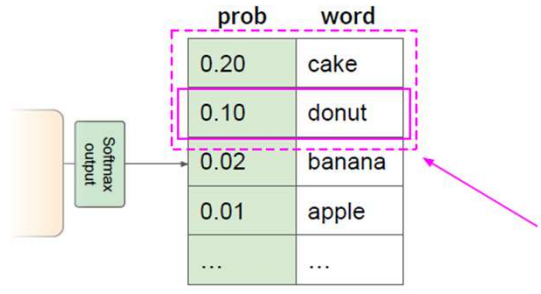
Generative config - top-k sampling



top-k: select an output from the top-k results after applying random-weighted strategy using the probabilities

15

Generative config - top-p sampling

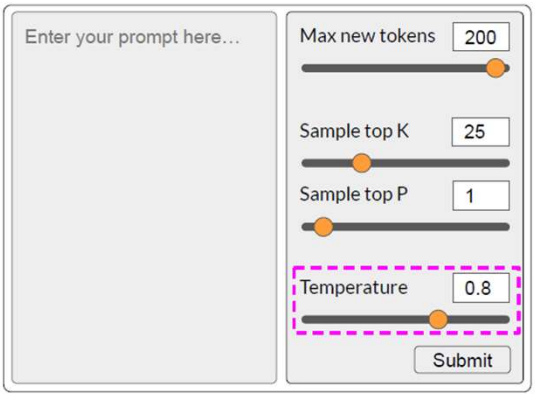


top-p: select an output using the random-weighted strategy with the top-ranked consecutive results by probability and with a cumulative probability <= *p*.

choose tokens until cumulative probability <= p

16

Generative configuration - temperature



Temperature

17

Generative config - temperature

To implement temperature scaling, we modify the standard Softmax function by dividing the raw scores (logits) by a temperature parameter T.

The probability P_i for a specific token i is calculated as:

$$P_i = \frac{\exp(z_i/T)}{\sum_{j=1}^{K} \exp(z_j/T)}$$

Variable Definitions

- z_i: The **logit** (raw score) for the i-th token in the vocabulary.
- T: The **Temperature** parameter.
- K: The total number of tokens in the model's vocabulary.
- $\exp$: The exponential function (e^x).

18

controls randomness

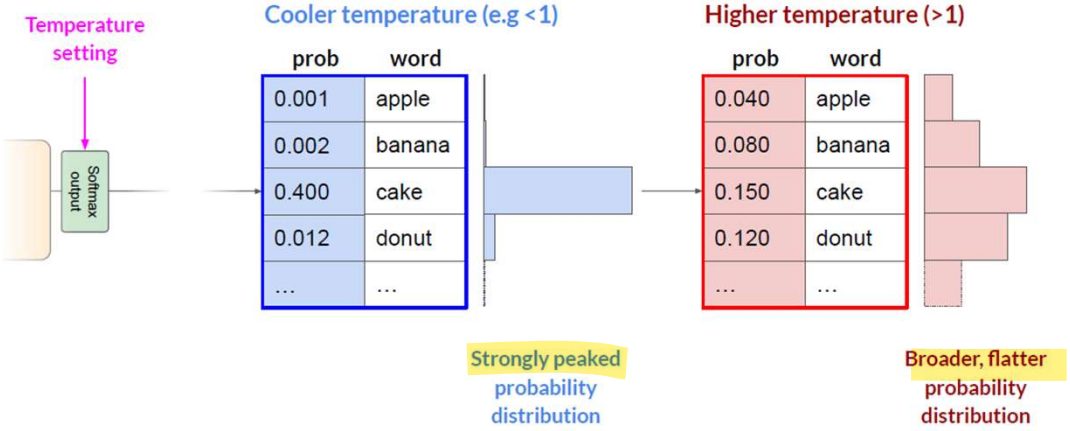
Generative config - temperature

How the Formula Affects Output

1. **When $T = 1$:** normal
   The formula is identical to the standard Softmax. The model's original confidence levels are preserved.
2. **When $T \to 0$ (Low Temperature):** T<1 - deterministic
   The differences between logits are amplified. The highest logit z_{max} dominates the fraction, making P_{max} approach **1.0** and all other probabilities approach **0**. This results in "**Greedy**" behavior.
3. **When $T > 1$ (High Temperature):** T>1- creative
   The logits are pushed closer together. The "gap" between the most likely and least likely word shrinks, creating a **flatter probability distribution**. This makes the model more likely to pick "surprising" words, which we perceive as creativity.

19

Generative config - temperature



20

Generative config - temperature

Temperature	Label	Effect on Model	Vibe
T<1.0	Cool	Increases the gap. The top choice becomes much more likely than the others.	Conservative: Focused, factual, but can be repetitive.
T=1.0	Neutral	The original distribution is preserved.	Balanced: The "intended" behavior of the model.
T>1.0	Hot	Decreases the gap. Low-probability words get a higher chance of being picked.	Creative: Diverse, unexpected, but prone to "hallucinations" or gibberish.

21

Huggingface Transformers

State-of-the-art pretrained models for inference and training

22

Huggingface Transformers

- Hugging Face Transformers is an open-source Python library that provides access to thousands of pre-trained Transformers models for natural language processing (NLP), computer vision, audio tasks, and more.
- It simplifies the process of implementing Transformer models by abstracting away the complexity of training or deploying models in lower level ML frameworks like PyTorch, TensorFlow and JAX.
- Transformers acts as the model-definition framework for state-of-the-art machine learning with text, computer vision, audio, video, and multimodal models, for both inference and training.
- Transformers works with **Python 3.10+**, and **PyTorch 2.4+**.

https://github.com/huggingface/transformers

23

Huggingface Transformers

24

How to use?

```
pip install transformers
```

HF workflow follows this 3-step pattern:


```
tokenizer = AutoTokenizer.from_pretrained("siebert/sentiment-roberta-large-english")
model     = AutoModelForSequenceClassification.from_pretrained(...)
outputs   = model(**tokenizer(text, return_tensors="pt"))
```

25

Tokenizers: From Text to Numbers

Three Ways to Load a Tokenizer

`DistilBertTokenizer`	Model-specific, Python	
`DistilBertTokenizerFast`	Model-specific, Rust (faster)	
`AutoTokenizer`	Auto-detects model type	✦ Recommended

26

Tokenizer: Step by Step

```
input_tokens = tokenizer.tokenize(input_str)
# → ["Hu", "##gging", "Face", "Transformers", "is", "great", "!"]

input_ids = tokenizer.convert_tokens_to_ids(input_tokens)
# → [20164, 10932, 10289, 25267, 1110, 1632, 106]

input_ids_special = cls + input_ids + sep
# → [101, 20164, ..., 106, 102]   ([CLS]...[SEP])

decoded = tokenizer.decode(input_ids_special)
# → "[CLS] Hugging Face Transformers is great! [SEP]"
```

- **Special Tokens** — [CLS] at start, [SEP] at end — model-dependent
- **Subword Tokenization** — "Hugging" → "Hu" + "##gging" — handles unknown words
- **Attention Mask** — 1 for real tokens, 0 for padding — tells model what to attend to

27

padding- ensures equal length of tokens

Padding, Truncation & Batching

```
model_inputs = tokenizer(
    ["Short text", "A much longer text..."],
    return_tensors="pt", padding=True, truncation=True
)
```

What happens:

Sentence 1:	[CLS]	Short	text	[SEP]	[PAD]	[PAD]	[PAD]
Sentence 2:	[CLS]	A	much	longer	text	...	[SEP]
Attn Mask:	1	1	1	1	0	0	0

← Sentence 1's attention mask (padding = 0)

28

Models: Architecture & Heads

Three Types of Transformer Models

Encoders
BERT, DistilBERT
Classification, NER, analysis

Decoders
GPT-2, LLaMA
Text generation

Enc-Dec
BART, T5
Summarization, translation

Task-Specific Model Heads

*Model	Base representations	*ForTokenClassification	NER, POS tagging
*ForMaskedLM	Fill in the blank	*ForQuestionAnswering	Extractive QA
*ForSequenceClassification	Sentiment, topic classification	*ForCausalLM	Text generation

29

Model Inference: It's Just PyTorch!

```
# Pass inputs via keyword args or dict unpacking
model_inputs = tokenizer(text, return_tensors="pt")
model_outputs = model(**model_inputs)

# Calculate loss normally
loss = F.cross_entropy(outputs.logits, labels)
loss.backward()  # Standard PyTorch!
```

Standard nn.Module	Use any PyTorch optimizer, scheduler, or loss function
Built-in Loss	Pass 'labels' kwarg and the model computes loss automatically
Attention & Hidden States	Set output_attentions=True and output_hidden_states=True

30

Fine-tuning: Loading Datasets

IMDB Dataset

25,000 train / 25,000 test reviews
Binary sentiment: POSITIVE / NEGATIVE
We use 128 train + 32 val (for speed)
Truncated to 50 tokens each

```
from datasets import load_dataset

ds = load_dataset("stanfordnlp/imdb")

# Tokenize the dataset
tokenized = ds.map(
  lambda ex: tokenizer(
    ex['text'], padding=True,
    truncation=True),
  batched=True)
```

Dataset Preprocessing Pipeline

1. Tokenize

Apply tokenizer to text with padding & truncation

2. Remove text

remove_columns(["text"]) — model only needs IDs

3. Rename label

rename_column("label", "labels") — HF convention

4. Set format

set_format("torch") — convert to PyTorch tensors

31

Fine-tuning: The Training Loop

Option A: PyTorch Training Loop

```
optimizer = AdamW(model.parameters(),
                  lr=5e-5)

for epoch in range(num_epochs):
  model.train()
  for batch in train_dataloader:
    output = model(**batch)
    output.loss.backward()
    optimizer.step()
    optimizer.zero_grad()

  model.eval()  # validation
  for batch in eval_dataloader:
    with torch.no_grad():
      output = model(**batch)
```

Option B: HF Trainer (recommended)

```
args = TrainingArguments(
  output_dir="checkpoints",
  per_device_train_batch_size=16,
  num_train_epochs=2,
  eval_strategy="epoch",
  learning_rate=2e-5)

trainer = Trainer(
  model=model,
  args=args,
  train_dataset=train_ds,
  eval_dataset=val_ds,
  compute_metrics=fn)

trainer.train()
```

32

HF Trainer: Callbacks & Evaluation

Callbacks

EarlyStoppingCallback
Stop training when validation loss plateaus

LoggingCallback (custom)
Log metrics to JSONL file on each logging step

Custom callbacks can hook into:
`on_log, on_epoch_end, on_evaluate, on_save, ...`

Evaluation

trainer.evaluate()
Returns evaluation metrics

trainer.predict(test_ds)
Returns predictions + metrics

compute_metrics function:
Custom metric calculation at eval time (accuracy, F1, etc.)

Recommended Starting Hyperparameters

Epochs	Learning Rate	Batch Size	Weight Decay	Warmup Steps
2–4	2e-5 or 5e-5	As large as fits GPU	0, 0.01, or 0.1	0, 100, or 500

33

Text Generation with GPT-2

```
from transformers import AutoModelForCausalLM, AutoTokenizer

tokenizer = AutoTokenizer.from_pretrained('gpt2')
model = AutoModelForCausalLM.from_pretrained('distilgpt2')

prompt = "Once upon a time"
tokenized = tokenizer(prompt, return_tensors="pt")

output = model.generate(**tokenized,
    max_length=50, do_sample=True, top_p=0.9)
```

Key Generation Parameters

max_length	Maximum number of tokens to generate
do_sample	Enable sampling (vs greedy decoding)
top_p	Nucleus sampling — only consider top-p probability mass
temperature	Controls randomness: lower = more deterministic
top_k	Only sample from top-k most likely tokens

34

Pipelines & Masked Language Modeling

Pipelines: One-Line Inference

```
sa = pipeline("sentiment-analysis",
  model="siebert/...")
sa("This movie is great!")
```

Available pipelines: sentiment-analysis, fill-mask, text-generation, question-answering, summarization, translation, NER, ...

Masked Language Modeling

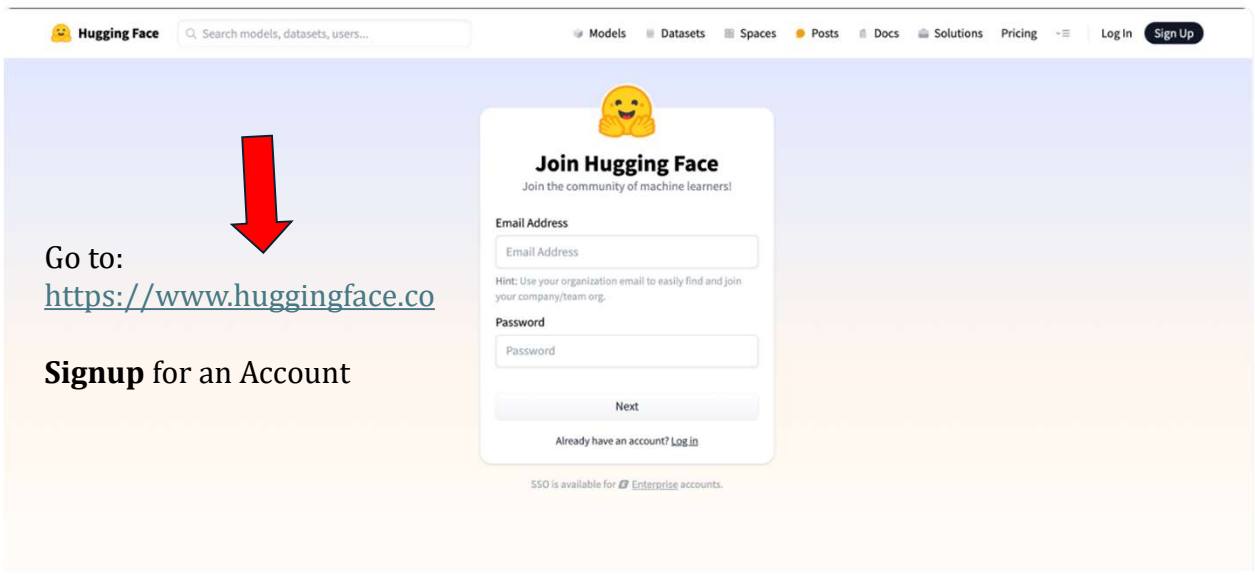
```
mlm =
pipeline("fill-mask","bert-base-cased")
mlm("I am [MASK] to learn!")
```

Top predictions:
→ excited (35.5%)
→ going (15.6%)
→ eager (7.9%)
→ here (3.6%)

Custom Datasets (Appendix 1 in notebook)

You can create custom PyTorch Datasets for HF models — just return tokenized dicts from __getitem__. See the E2E Dataset example in the notebook for encoder-decoder tasks.

35 pipelines- simplify inference

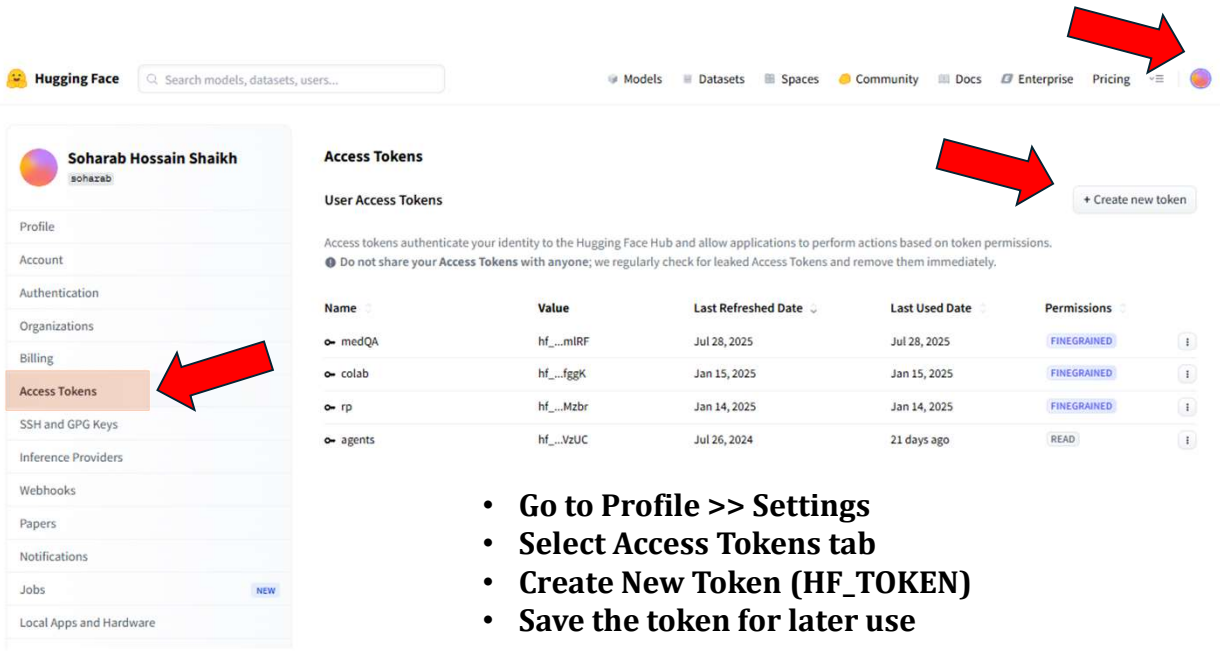


Go to:
https://www.huggingface.co

Signup for an Account

36

HF_TOKEN- used for authentication



- **Go to Profile >> Settings**
- **Select Access Tokens tab**
- **Create New Token (HF_TOKEN)**
- **Save the token for later use**

37

Why should you use Transformers?

1. Easy-to-use state-of-the-art models:

- High performance on natural language understanding & generation, computer vision, audio, video, and multimodal tasks.
- Low barrier to entry for researchers, engineers, and developers.
- Few user-facing abstractions with just three classes to learn.
- A unified API for using all provided pretrained models.

2. Lower compute costs, smaller carbon footprint:

- Share trained models instead of training from scratch.
- Reduce compute time and production costs.
- Dozens of model architectures with 1M+ pretrained checkpoints across all modalities.

3. Choose the right framework for every part of a model's lifetime:

- Train state-of-the-art models in 3 lines of code.
- Move a single model between PyTorch/JAX/TF2.0 frameworks at will.
- Pick the right framework for training, evaluation, and production.

4. Easily customize a model or an example to your needs:

- We provide examples for each architecture to reproduce the results published by its original authors.
- Model internals are exposed as consistently as possible.
- Model files can be used independently of the library for quick experiments.

38

References

1. **DeepLearning.ai course: Generative AI and LLMs with AWS**
https://www.deeplearning.ai/courses/generative-ai-with-llms/

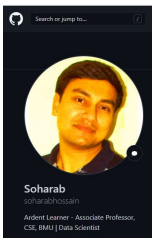
2. **Huggingface Course on Youtube (Playlist)**
https://www.youtube.com/watch?v=00GKzGyWFEs&list=PLo2EIpI_JMQvWfQndUesu0nPBAtZ9gP1o

3. **Total noob's intro to Hugging Face Transformers**
https://huggingface.co/blog/noob_intro_transformers

4. **Transformers Notebooks**
https://huggingface.co/docs/transformers/en/notebooks
https://github.com/huggingface/notebooks

39

40



Thank You

Class material will be uploaded here:
https://github.com/soharabhossain/GenAI_n_LLMs